

Bioinformatics of Exoprotocells

LIFEORIG working note

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1 Purpose

This note connects the Titan protocell framework to a bioinformatics program. The existing Titan notes define a methane-pool environment, a reaction network, phase allocation, dense organic phases, and interface-stabilized protocell candidates. The next computational question is how to analyze the resulting population once many candidate compartments exist.

The central proposal is that an exoprotocell does not need a DNA or RNA genome in order to generate an informatic state at the population level. The protocell itself is first a dynamical physical-chemical object. Its state evolves by internal reaction kinetics, transport across the boundary, structural change of the compartment, and coupling to the local environment. A useful abstract state is:

$$\mathcal{P}_k = (\mathbf{C}^{P_k}, \mathbf{u}^{P_k}, \mathbf{z}^{P_k}, \mathbf{E}(z_k, t), \mathcal{H}_k), \quad (1)$$

where $\mathbf{C}^{P_k}$ is the internal chemical composition, $\mathbf{u}^{P_k}$ is the boundary/interface structure and permeability state, $\mathbf{z}^{P_k}$ contains physical object variables such as radius, volume, and surface area, $\mathbf{E}(z_k, t)$ is the local external environment at the protocell position, and $\mathcal{H}_k$ is the object history.

The active part of the chemical network is not assumed to be fixed inside the protocell. It is an emergent subnetwork selected by the current state:

$$\mathcal{R}_{\text{active}}^{P_k}(t) = \mathcal{A}(\mathbf{C}^{P_k}(t), \mathbf{u}^{P_k}(t), \mathbf{z}^{P_k}(t), \mathbf{E}(z_k, t)). \quad (2)$$

Bioinformatics enters after this dynamical step: it analyzes the distribution of protocell states, structures, active subnetworks, and lineages across a population.

2 Relation to the Titan Theory

The Titan design note separates chemical identity from physical phase allocation:

$$\text{chemical species classification} \neq \text{physical phase allocation}. \quad (3)$$

This distinction is also the starting point for exoprotocell bioinformatics. The relevant unit is not only a molecule sequence. It is a coupled object:

$$\text{species} + \text{phase} + \text{reaction role} + \text{retention} + \text{history}. \quad (4)$$

The Titan model promotes compartments by local threshold rules:

$$C_{\text{coa}}(z_j, t) \geq C_{\text{coa}}^{\text{crit}} \Rightarrow \text{dense organic object}, \quad (5)$$

$$C_I(z_j, t) \geq C_I^{\text{crit}} \Rightarrow \text{interface-stabilized protocell candidate}. \quad (6)$$

Bioinformatics begins after this promotion step. Each promoted object becomes a sample in a population dataset.

3 Dynamical State and Informatic State

For an ordinary cell, bioinformatics often starts from sequence data. For a Titan exoprotozell candidate, the first object is instead the dynamical state:

$$\frac{d\mathbf{C}^{P_k}}{dt} = \mathbf{S}\mathbf{r}^{P_k} + \mathbf{J}_{\text{in}}^{P_k} - \mathbf{J}_{\text{out}}^{P_k} - \mathbf{D}^{P_k}, \quad (7)$$

where $\mathbf{S}$ is the stoichiometric matrix, $\mathbf{r}^{P_k}$ is the vector of internal reaction rates, $\mathbf{J}_{\text{in}}^{P_k}$ and $\mathbf{J}_{\text{out}}^{P_k}$ are exchange fluxes with the local environment, and $\mathbf{D}^{P_k}$ represents degradation, leakage, burial, or photochemical damage.

The exchange terms depend on both the protocell structure and the local environment:

$$J_{\text{in},i}^{P_k} = \frac{A_k}{V_k} P_i(\mathbf{u}^{P_k}, \mathbf{E}(z_k, t)) C_i^{\text{ext}}(z_k, t), \quad (8)$$

$$J_{\text{out},i}^{P_k} = \frac{A_k}{V_k} P_i(\mathbf{u}^{P_k}, \mathbf{E}(z_k, t)) C_i^{P_k}(t). \quad (9)$$

Thus two protocells with the same internal composition can diverge if they have different shell structures, surface coverage, permeability profiles, birth positions, or local chemical environments.

The informatic state is not an additional physical state inside the protocell. It is a feature representation extracted from the dynamical state and its history:

$$\mathbf{x}_k = [\mathbf{c}_k, \mathbf{p}_k, \mathbf{a}_k, \mathbf{g}_k, \mathbf{s}_k, \mathbf{h}_k]. \quad (10)$$

Block	Meaning
$\mathbf{c}_k$	internal composition vector for protocell candidate k
$\mathbf{p}_k$	phase and localization vector: internal, interface, adsorbed, dense, external-nearby
$\mathbf{a}_k$	activity vector: active reactions, fluxes, catalytic factors, exchange rates
$\mathbf{g}_k$	graph features of the currently active subnetwork
$\mathbf{s}_k$	structural vector: shell composition, permeability, radius, surface area, retention times
$\mathbf{h}_k$	history features: birth position, parent object, exposure, division, fusion, loss

The object to track is therefore not a private genome for each protocell. It is an effective organization that constrains which reactions from the shared network are expressed by the protocell state:

$$O_k^{\text{eff}}(t) = \left(\mathcal{R}_{\text{active}}^{P_k}, \mathcal{C}_{\text{catalysts}}^{P_k}, \mathbf{P}^{P_k}, \mathbf{u}^{P_k}, \mathbf{E}(z_k, t) \right). \quad (11)$$

Here $\mathbf{P}^{P_k}$ is the permeability profile. The boundary is included because, for non-aqueous protocells, chemical access and retention are part of the effective inherited organization.

4 Polymer-Generated Catalytic Diversity

The simulation has one global reaction network. Let

$$\mathcal{R} = \{r_1, \dots, r_M\}, \quad \mathcal{S} = \{s_1, \dots, s_N\}. \quad (12)$$

Every protocell uses the same $\mathcal{R}$ and $\mathcal{S}$. Diversity is not introduced by giving each protocell a different network. Diversity is introduced by allowing different protocells to develop different internal polymer populations.

Let the monomer alphabet be

$$\mathcal{M} = \{m_1, \dots, m_A\}. \quad (13)$$

A polymer sequence is

$$\pi_a = (\sigma_1, \dots, \sigma_{L_a}), \quad \sigma_j \in \mathcal{M}, \quad (14)$$

where L_a is the chain length. The number of possible sequences of length L is

$$|\Pi_L| = A^L. \quad (15)$$

Thus longer polymers create a combinatorial space of possible structures. Most sequences may be inactive, but some sequences can fold, aggregate, adsorb, or expose chemical groups in ways that alter reaction rates.

For protocell k , define the internal polymer copy number:

$$N_{a,k}^{\text{poly}}(t) = \text{number of copies of polymer sequence } \pi_a \text{ inside protocell } k. \quad (16)$$

The polymer population vector is

$$\mathbf{N}_k^{\text{poly}}(t) = (N_{1,k}^{\text{poly}}, \dots, N_{N_{\text{poly}},k}^{\text{poly}}). \quad (17)$$

The key biochemical object is the polymer-to-reaction catalytic coefficient:

$$\Gamma_{ar} = \text{catalytic strength of polymer } \pi_a \text{ on reaction } r. \quad (18)$$

This coefficient can be zero for most polymer-reaction pairs. It can depend on sequence length, composition, conformation, phase localization, and solvent compatibility. In a first model it can be assigned statistically:

$$\Gamma_{ar} \sim P_{\Gamma}(L_a, \chi_a, r), \quad (19)$$

where χ_a is a sequence descriptor such as nitrile fraction, hydrophobic fraction, charge-like polarity class, or HCN-polymer motif count.

The effective catalytic activity of reaction r inside protocell k is then

$$\Gamma_{r,k}^{\text{eff}}(t) = \Gamma_r^0 + \sum_a \Gamma_{ar} \frac{N_{a,k}^{\text{poly}}(t)}{V_k(t)} \theta_{a,k}(t), \quad (20)$$

where Γ_r^0 is the background uncatalyzed or externally catalyzed contribution, V_k is protocell volume, and $\theta_{a,k}$ is the fraction of polymer π_a in an active location or conformation.

The reaction rate inside protocell k can then be written as

$$R_{r,k}(t) = \Gamma_{r,k}^{\text{eff}}(t) F_r(\mathbf{E}(z_k, t)) \prod_i \left[C_i^{P_k}(t) \right]^{\nu_{ir}}, \quad (21)$$

where $F_r(\mathbf{E})$ is the extrinsic environmental factor and $\Gamma_{r,k}^{\text{eff}}$ is the intrinsic protocell catalytic factor. This split is essential:

$$\text{environmental forcing} \neq \text{intrinsic protocell diversity}. \quad (22)$$

A protocell-specific active reaction set can now be defined without changing the global network:

$$\mathcal{R}_{\text{active}}^{P_k}(t) = \{r \in \mathcal{R} : R_{r,k}(t) > R_{\min}\}. \quad (23)$$

The active set differs because $\mathbf{N}_k^{\text{poly}}$, $\theta_{a,k}$, and $\Gamma_{r,k}^{\text{eff}}$ differ, not because $\mathcal{R}$ differs.

5 Measuring Protocell Diversity

The intrinsic diversity variable should be

$$X_k(t) = \left(\mathbf{N}_k^{\text{poly}}(t), \mathbf{\Gamma}_k^{\text{eff}}(t), \mathbf{C}^{P_k}(t), \mathbf{P}_k(t), \boldsymbol{\theta}_k(t) \right), \quad (24)$$

where

$$\mathbf{\Gamma}_k^{\text{eff}}(t) = \left(\Gamma_{1,k}^{\text{eff}}, \dots, \Gamma_{M,k}^{\text{eff}} \right), \quad (25)$$

$$\mathbf{P}_k(t) = (P_{1,k}, \dots, P_{N,k}), \quad (26)$$

$$\boldsymbol{\theta}_k(t) = (\theta_{1,k}, \dots, \theta_{N_{\text{poly}},k}). \quad (27)$$

Here $\mathbf{P}_k$ is the permeability vector for species transport across the protocell boundary. The vector $\boldsymbol{\theta}_k$ records which polymers are actually folded, surface-bound, dense-phase localized, or otherwise catalytically available.

A concrete distance between protocells is:

$$\begin{aligned} d_{\text{proto}}(k, l) = & w_{\text{poly}} \frac{\|\mathbf{N}_k^{\text{poly}} - \mathbf{N}_l^{\text{poly}}\|_1}{N_k^{\text{poly}} + N_l^{\text{poly}} + \epsilon} \\ & + w_{\Gamma} \|\mathbf{\Gamma}_k^{\text{eff}} - \mathbf{\Gamma}_l^{\text{eff}}\|_2 + w_C \|\mathbf{C}^{P_k} - \mathbf{C}^{P_l}\|_1 \\ & + w_P \|\mathbf{P}_k - \mathbf{P}_l\|_2 + w_{\theta} \|\boldsymbol{\theta}_k - \boldsymbol{\theta}_l\|_1. \end{aligned} \quad (28)$$

This is the object that should be clustered in a bioinformatics analysis. It measures intrinsic protocell differences. Local environment can be stored separately as metadata:

$$\mathbf{E}_k(t) = \mathbf{E}(z_k, t). \quad (29)$$

The environment may change measured activity, but it should not be confused with protocell identity.

6 Exoprotocell Quasispecies

The population-level object is the quasispecies. In this framework, a quasispecies is not a cloud of nucleic-acid sequences. It is a cloud of protocell states whose main intrinsic diversity comes from retained polymer populations and the catalytic matrices generated by those polymers:

$$\mathcal{Q}(t) = \left\{ \left(\mathbf{N}_k^{\text{poly}}, \mathbf{\Gamma}_k^{\text{eff}}, \mathbf{C}^{P_k}, \mathbf{P}_k, \boldsymbol{\theta}_k \right) \right\}_{k=1}^{N_P(t)}. \quad (30)$$

The local environment remains important, but it is not the definition of protocell type. A protocell at the UV-exposed surface and a protocell attached to a mineral surface may have different observed rates because $\mathbf{E}(z_k, t)$ is different. That is environmental heterogeneity. Protocell diversity requires differences in $X_k(t)$ itself: different retained polymers, different polymer conformational availability, different boundary permeability, and therefore different $\mathbf{\Gamma}_k^{\text{eff}}$.

Let $\phi_{\alpha}(t)$ be a coarse-grained protocell type, obtained by clustering records $\mathbf{x}_k$ into an equivalence class:

$$\begin{aligned} \phi_{\alpha} = & \left[\text{polymer-population fingerprint, effective-catalysis fingerprint,} \right. \\ & \left. \text{internal-composition fingerprint, permeability fingerprint, active-conformation fingerprint} \right]_{\alpha}. \end{aligned} \quad (31)$$

The quasispecies distribution is then

$$q_\alpha(t) = \frac{N_\alpha(t)}{\sum_\beta N_\beta(t)}. \quad (32)$$

A useful first dynamical form is:

$$\frac{dq_\alpha}{dt} = \sum_\beta W_{\alpha\beta} q_\beta - \bar{W} q_\alpha, \quad (33)$$

where $W_{\alpha\beta}$ is an effective transition-selection operator. It includes polymer growth, polymer cleavage, polymer loss, sequence-dependent catalysis, protocell growth, fragmentation, fusion, and boundary modification. Environmental relocation can change the observed rates, but a transition between intrinsic types requires a change in X_k . The mean population persistence is:

$$\bar{W} = \sum_{\alpha,\beta} W_{\alpha\beta} q_\beta. \quad (34)$$

In this sense, mutation is generalized. A protocell type can change because its polymer population changes:

$$\mathbf{N}_k^{\text{poly}} \rightarrow \mathbf{N}_k^{\text{poly}} + \Delta \mathbf{N}_k^{\text{poly}}, \quad (35)$$

which induces

$$\mathbf{\Gamma}_k^{\text{eff}} \rightarrow \mathbf{\Gamma}_k^{\text{eff}} + \Delta \mathbf{\Gamma}_k^{\text{eff}}. \quad (36)$$

The formal reaction network remains fixed. The quasispecies is the population distribution over polymer-generated catalytic states.

7 Exoprotocell Omics

The following analogs are useful for implementation and analysis.

Earth bioinformatics term	Exoprotocell analog	Data object
Genome-like state	retained polymer population inside the protocell	vector
Transcriptome-like state	reaction-rate vector induced by the polymer population	vector
Proteome-like state	catalytically active polymers, conformations, and surface sites	set/vector
Metabolome	internal chemical composition	vector
Lipidome	interface or shell composition	vector
Microbiome	local population of interacting protocells	graph
Phylogeny	lineage of birth, fusion, fragmentation, inheritance	tree/DAG

These labels should be used carefully. They are computational analogies, not claims that Titan protocells possess terrestrial biology.

8 Population Bioinformatics

A simulation output should be convertible into a population table:

$$\mathcal{D}(t) = \{\mathbf{x}_1(t), \dots, \mathbf{x}_{N_P(t)}(t)\}. \quad (37)$$

The minimal columns should be:

$$\begin{aligned} \text{id, parent_id, birth_time, birth_z, radius, interface_type,} \\ \text{polymer_fingerprint, gamma_eff_fingerprint,} \\ \text{permeability_fingerprint, retention_score.} \end{aligned} \quad (38)$$

This enables standard population analyses:

- clustering of protocell states,
- trajectory analysis through chemical state space,
- detection of recurrent polymer-generated catalytic motifs,
- lineage reconstruction after fragmentation or division-like events,
- survival analysis of compartment lifetimes,
- association between interface composition and persistence.

9 Lineages Without Genes

In early exoprotocell simulations, inheritance may occur through partial material continuity rather than sequence copying. If protocell k fragments into daughters a and b , inheritance can be measured by composition overlap:

$$H_{a|k} = \frac{\sum_i \min(C_i^{P_a}, C_i^{P_k})}{\sum_i C_i^{P_a}}. \quad (39)$$

Polymer inheritance can be measured by polymer population overlap:

$$J_{\text{poly}}(a, k) = \frac{\sum_p \min(N_{p,a}^{\text{poly}}, N_{p,k}^{\text{poly}})}{\sum_p \max(N_{p,a}^{\text{poly}}, N_{p,k}^{\text{poly}})}. \quad (40)$$

Effective catalytic inheritance can be measured by the similarity of the induced catalytic matrices:

$$J_{\Gamma}(a, k) = 1 - \frac{\|\mathbf{\Gamma}_a^{\text{eff}} - \mathbf{\Gamma}_k^{\text{eff}}\|_2}{\|\mathbf{\Gamma}_a^{\text{eff}}\|_2 + \|\mathbf{\Gamma}_k^{\text{eff}}\|_2 + \epsilon}. \quad (41)$$

The lineage graph is then a directed acyclic graph:

$$\mathcal{L} = (V, E), \quad (42)$$

where vertices are protocell objects and edges record division, fragmentation, fusion, or persistent material transfer.

10 Feature Engineering for Titan

For Titan specifically, the most important features are not fast growth features. They are persistence features:

$$\begin{aligned} S_{\text{persist},k} = & w_1 \tau_{\text{survival},k} + w_2 \tau_{\text{retention},k} + w_3 \theta_{I,k} + w_4 M_{\text{polymer},k} \\ & + w_5 R_{\text{catalytic},k} - w_6 D_{\text{photochemical},k} - w_7 L_{\text{sediment},k}. \end{aligned} \quad (43)$$

The corresponding bioinformatics features should include:

- nitrile-rich composition fraction,
- HCN-polymer or tholin-like polymer fraction,
- interface-forming species fraction,
- polymer sequence-length distribution,
- polymer sequence diversity or Shannon entropy,
- effective catalytic matrix $\mathbf{\Gamma}_k^{\text{eff}}$,
- catalytic closure score induced by retained polymers,
- RAF or autocatalytic subset size,
- retention time distribution over key species,
- phase entropy, measuring whether material is concentrated or dispersed,
- environmental birth context: surface, sediment, adsorbed layer, or dense organic phase.

One useful phase entropy is

$$H_{\text{phase},k} = - \sum_m q_{m,k} \log q_{m,k}, \quad (44)$$

where $q_{m,k}$ is the fraction of protocell-associated material in phase m . Low entropy may indicate strong compartmentalization; high entropy may indicate diffuse material with weak protocell identity.

11 Implementation Proposal

The implementation should not replace the chemistry or dynamics layer. It should be an analysis layer that consumes simulation state snapshots.

11.1 New data objects

Recommended first objects:

$$\begin{aligned} &\text{ExoprotocellRecord, ExoprotocellPopulation, PolymerPopulation,} \\ &\text{EffectiveCatalyticMatrix, LineageGraph.} \end{aligned} \quad (45)$$

Object	Responsibility
ExoprotocellRecord	store one protocell candidate snapshot
ExoprotocellPopulation	store all protocell snapshots at time t
PolymerPopulation	store polymer sequence counts $N_{a,k}^{\text{poly}}$ for one protocell
EffectiveCatalyticMatrix	compute $\mathbf{\Gamma}_k^{\text{eff}}$ from retained polymers
LineageGraph	record parent, daughter, fusion, and transfer relations

11.2 Analysis functions

Recommended first functions:

```
fingerprint_composition(record),
fingerprint_polymer_population(record),
compute_effective_catalytic_matrix(record),
compute_closure_score(record),
compute_retention_score(record),
compare_records(record_a, record_b),
build_lineage(records).
```

(46)

The output should be plain tables and sparse matrices first. More elaborate visualization can come later.

12 Connection to Current LIFEORIG Code

The existing code already has several useful anchors:

- `MoleculeModel`, `BinaryPolymer`, `MultiPolymer`, and `ReferenceMolecule` can define chemical identity records.
- `ChemicalNetworkParser` can provide parsed species and reaction records.
- ACF/RAF-style network routines can provide closure and autocatalytic subset features.
- Titan environment parsers provide the environmental context needed for birth and persistence features.
- The future `TitanPhaseAllocator` can provide phase labels and phase inventories.

The key design rule is:

bioinformatics layer = read-only analysis of simulated protocell states. (47)

It should not own the chemistry, the environmental dynamics, or the phase transition rules.

13 Minimal First Milestone

The first useful milestone is a post-processing pipeline:

1. Run a Titan methane-pool simulation.
2. Save protocell candidate snapshots at selected times.
3. Convert each candidate into an `ExoprotocellRecord`.
4. Build polymer-population fingerprints and effective-catalytic-matrix fingerprints.
5. Cluster candidates by $d_{\text{proto}}(k, l)$.
6. Build a lineage graph from parent-child or fragmentation metadata.
7. Report persistence-associated features.

The scientific output is not a claim of life. It is a map of which chemical organizations persist, recur, and transmit structure under Titan-like constraints.

14 References and Local Inputs

Local LIFEORIG notes and code

- `doc/TITAN/titan_methane_pool_protocell_design.tex`
- `doc/TITAN/titan_catalysis_framework_latex.pdf`
- `doc/TITAN/titan_protocell_framework.pdf`
- `src/chemical_types/molecule_model.py`
- `src/input_data/chemical_network_parser.py`
- `src/RAF/RAF_engine.py`

Literature for polymer-generated protocell diversity

- Walker, S. I., Grover, M. A., and Hud, N. V. Universal sequence replication, reversible polymerization and early functional biopolymers: A model for the initiation of prebiotic sequence evolution. 2012. <https://arxiv.org/abs/1203.4625>
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